

RGMC2026 H2R Handover Track Solution

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Abstract—Achieving safe and natural human-to-robot handovers remains a critical challenge in human-robot interaction, particularly when utilizing rigid, high-payload industrial hardware where safety margins are minimal. The Human-to-Robot Handover track at the Robotic Grasping and Manipulation Competition (RGMC) 2026 benchmarks autonomous systems on their ability to safely receive unknown containers from humans under zero-shot constraints. We present a robust, highly-tunable, distributed industrial system based on a rigid parallel-jaw gripper and multi-modal sensing, combining RGB-D perception, force-torque feedback, and custom tactile sensing. In the RGMC 2026 competition, the system achieved the highest qualification score of 71.41 points and a 75% success rate in the final evaluation across diverse lighting and network conditions. The core contribution is a human-aware grasping pipeline that combines diffusion-based grasp generation with geometric and kinematic filtering to explicitly avoid collisions with human fingers while maintaining high manipulability. In addition, we incorporate tactile feedback to improve grasp stability during object acquisition. The system operates in real time and requires minimal deployment effort, enabling rapid on-site setup.

I. INTRODUCTION

Human-to-robot (H2R) handover requires robots to safely acquire objects from a human while adapting to unpredictable human motion. Compared to conventional pick-and-place tasks, handovers combine dynamic perception, reactive motion planning, and safe physical interaction in close proximity to a human. These challenges become more pronounced when using rigid industrial manipulators and parallel-jaw grippers, where grasp planning and motion execution must remain both responsive and collision-aware.

To bridge this gap between industrial rigidity and safe human collaboration, we present a robust, distributed, and sensor-rich robotic system designed explicitly for the Human-to-Robot Handover track of the 11th Robotic Grasping and Manipulation Competition (RGMC) at ICRA 2026. Utilizing a standard Universal Robots manipulator equipped with a Robotiq parallel-jaw gripper, our architecture integrates heterogeneous sensing modalities—specifically, dual static RGB-D cameras, wrist-mounted force-torque (FT) sensing, and custom-built, high-sensitivity tactile finger arrays—to enhance situational awareness during the interaction.

The primary contribution is a human-aware grasp planning pipeline that combines diffusion-based grasp generation with application-specific geometric and kinematic filtering. Candidate grasps generated by *GraspGen* model [1] are refined using constraints on object orientation, robot manipulability, and explicit filtering of grasps that intersect or approach the

human hand. The perception pipeline further separates fast object tracking from high-accuracy geometric reconstruction using specialized vision models, allowing efficient real-time tracking without sacrificing grasp quality.

II. HARDWARE AND SETUP

We utilize a Universal Robots UR7e robot arm, chosen to provide the necessary reach and payload capabilities. A Robotiq 2F-140 parallel-jaw gripper is selected as the end-effector due to its large stroke, allowing us to plan for variable-sized objects as well as enabling effortless integration with the grasp generation model.

A. Perception and Sensing

The visual workspace is monitored by two Intel RealSense D435i cameras, providing steady RGB-D streams at 640×480 resolution and 30 FPS. In order to obtain the transformations from the camera frame to the robot's base frame, we use the *easy_handeye_2* library [2]. A calibration board attached to the robot end-effector is observed from multiple poses, allowing the camera-to-base transformation to be estimated. Using 20–30 calibration samples, the resulting registration accuracy is approximately 0.5–2 cm. To enhance motion control with physical-world interaction, the end-effector setup is augmented with a Robotiq FT 300-S Force-Torque Sensor for touchdown detection and custom-built, barometric-pressure-sensor-based tactile sensors to ensure delicate and accurate grasping. The tactile sensors are equipped with a grid of 7 taxels and possess high sensitivity, enabling the detection of highly deformable objects. These sensors stream data over Bluetooth Low Energy (BLE) at up to 60 FPS per finger.

B. Computing Infrastructure

To guarantee strict real-time execution, the computational workload is bifurcated across two workstations, connected via a 2.5 Gbps Ethernet link:

- **Vision Pipeline Workstation:** Equipped with an AMD Ryzen 9 9950X 16-Core Processor and an NVIDIA RTX PRO 6000 Blackwell GPU (96 GB VRAM), this workstation handles the visual perception system.
- **Motion Planning Workstation:** Powered by an AMD Ryzen 9 5950X 16-Core Processor, this system is exclusively reserved for motion generation. It features a dedicated 1 Gbps hardware Ethernet connection to the UR7e controller.

C. System Environment

The entire software stack is deployed on **Ubuntu Linux 22.04 LTS** utilizing ROS2 Humble. Eclipse Cyclone DDS is used as ROS2 communication backbone.

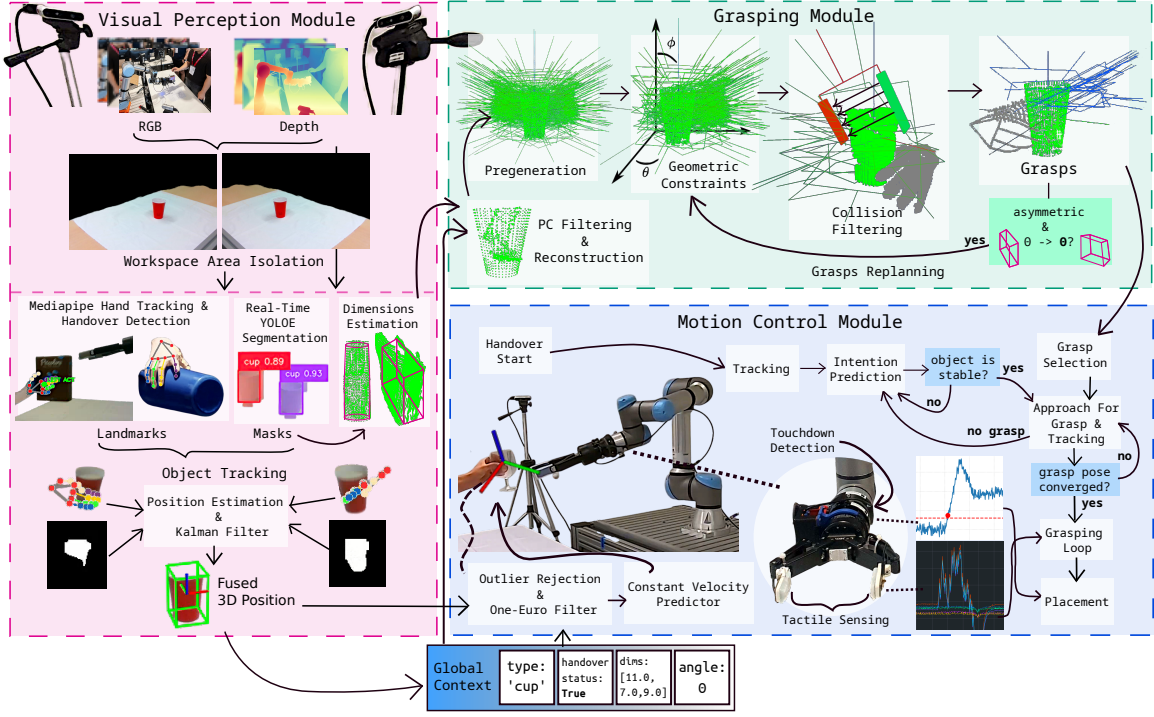


Fig. 1: System architecture and operational pipeline for the human-to-robot handover task.

III. SYSTEM ARCHITECTURE

The system is implemented as a distributed ROS2 architecture designed to decouple high-frequency motion control from perception and maximize data sharing between perception components. It is organized into three main modules: the *Visual Perception Module*, the *Grasping Module*, and the *Motion Control Module* (Figure 1).

A. Visual Perception Module

The perception module processes dual RGB-D camera streams to estimate object and human hand states in the shared workspace. It starts with a depth-based background filter that removes extraneous RGB pixels to isolate the active workspace and discard environmental noise. Time synchronization across asynchronous ROS2 nodes is handled using timestamp alignment. The module provides estimates of object position, maintains a global workspace representation, and triggers higher-level interaction events. Perception outputs are shared with downstream modules through a centralized message interface, enabling real-time updates for grasping and motion planning. To improve efficiency, perception is split into parallel sub-processes that independently contribute to data fusion.

1) *Real-Time Object Segmentation*: Object segmentation is performed using a real-time open-vocabulary detector (YOLOE [3]), which generates segmentation masks from both camera views. The model provides robust performance under varying lighting conditions and does not require environment-specific pretraining when embedded within our pipeline. The raw outputs can occasionally suffer from intermittent label misidentification and coarse boundary definitions, necessitating downstream filtering. The resulting segmentation stream

is used as the primary input for pose estimation. The detector is conditioned on the following semantic sets:

Listing 1: Semantic container categorization priors.

```
symmetric_priors = ["cup", "glass", "bottle",
                    "mug", "can", "bowl", "plate"]
asymmetric_priors = ["box", "food box", "food
                     container", "food package", "bag", "food
                     bag"]
```

2) *Hand Tracking*: Hand tracking is performed using MediaPipe [4], which provides real-time 2D hand landmark detection. Handover events are triggered based on spatial proximity between the detected hand and the tracked object. A recency-weighted object reference point is computed from the last $N = 6$ observations:

$$\mathbf{r}(t) = \sum_{i=1}^N w_i \cdot \mathbf{p}_i, \quad w_i = \frac{i}{\sum_{j=1}^N j}$$

For each detected hand with landmarks $\mathcal{L}_h = \{l_1, \dots, l_{21}\}$, the minimum distance to the object reference is:

$$d_h = \min_{l \in \mathcal{L}_h} \text{dist}(l, \mathbf{r})$$

A handover event is triggered when $d_h < \tau$ with $\tau = 5$ pixels. Once triggered, this event is broadcast as part of the global context, and the tracked hand landmarks are integrated into the object tracking pipeline.

3) *Object Tracking*: This submodule implements dynamic object tracking and sensor fusion for robust 3D object localization. Object tracking is achieved by projecting the YOLOE segmentation masks onto the registered depth images from both cameras. The resulting spatial data is transformed into the robot base frame and fused, achieving a consistent 30 FPS

tracking loop under stable detection conditions. To guarantee robustness against segmentation failures or severe hand-object occlusion during physical exchange, the system dynamically incorporates MediaPipe's 2D hand landmarks as a secondary tracking modality. We apply Kalman filtering, outlier detection, and dynamic hand-object offset calculations to maintain a stable, fused trajectory even when the target object is temporarily obscured.

4) *Dimensions Estimator*: The dimensions estimator reconstructs object geometry from dual RGB-D camera observations. It constructs the global context containing object dimensions, initial orientation angle and container type. We categorize the container type using the semantic sets (Listing 1). The object type consistent with these sets is obtained from the locally deployed Vision-Language Model (VLM) (Qwen3-VL-8B) [5] for the most objective scene assessment.

To obtain dimensions, depth images and corresponding segmentation masks are back-projected into 3D point clouds using a pinhole model and transformed into the robot base frame using the calibrated camera extrinsics. Noise is reduced using statistical outlier removal ($k = 20, \sigma = 1$) for point cloud filtering, after which geometry is estimated using different strategies depending on object symmetry.

Asymmetrical objects: For box-like objects, an oriented bounding box (OBB) is fitted to the fused point cloud to extract principal dimensions and orientation. The resulting representation provides object height, width and yaw angle in the object frame.

Symmetrical objects: For rotationally symmetric or weakly structured objects, dimensions are estimated using robust image- and depth-based projections across multiple viewpoints. Height is derived from depth statistics, while horizontal extents are computed from projected mask boundaries.

We utilize SAM3 (Segment Anything Model) [6], prompted by the label from YOLOE, to extract highly precise segmentation masks for this task. In cases where the initial label failed producing a valid mask, the VLM's output is used as a fallback prompt. The segmentation masks are shared with the Grasping module.

B. Grasping Module

The grasping module generates kinematically feasible and human-aware grasp configurations. The module splits execution into a *Stage 1 (Object Pregeneration)* phase, a *Stage 2 (Hand Filtering)* phase, and an asymmetric container-specific *Stage 3 (Grasp Replanning)* phase. The hand point cloud is constructed by deprojecting the region enclosing hand landmarks if available; SAM3 segmentation is used as a fallback. **Stage 1**: The pipeline starts by pregenerating the grasps while the target object rests on the table. To condition the *GraspGen* model for higher synthesis quality particularly for poorly observed objects, the system conditionally activates a lightweight proxy reconstruction based on the estimated object dimensions (Section III-A4). The generated grasp set is filtered using geometric constraints to ensure feasibility:

- 1) *Adaptive Spherical Constraints*: Grasps are evaluated based on their approach angles (azimuth ϕ and elevation θ) in a spherical coordinate system.

- 2) *Roll Isomorphism Filtering*: Grasp candidates are flattened to a horizontal orientation relative to the world- Y axis to enforce level finger alignment. Poses are eliminated if their localized roll rotation about the gripper Z -axis exceeds a strict safety threshold ($\leq 20.0^\circ$).

Stage 2: Upon detection of a handover event, the system switches to collision filtering. A voxelized representation of the human hand holding the object is constructed from estimated hand point cloud. Each candidate grasp is evaluated using a fast ray-casting-based collision check between the gripper fingers across the maximum aperture. Grasps that intersect the hand volume are rejected. If no collision-free grasp remains, the system selects the candidate with minimal hand-overlap as a fallback. The final grasp set is ranked and reduced to a top- k subset, biased toward grasp regions closer to gripper's approach vector. The selected grasps, together with required finger opening widths, are passed to the Motion Control module.

Stage 3: If object orientation changes during the handover, the pregenerated grasp set is rotated accordingly, and the same filtering pipeline is re-applied. Newly selected grasps are utilized for motion replanning.

C. Motion Control

The Motion Control module executes the handover by combining visual servoing, trajectory generation, and sensor-based interaction control. Motion is realized with a *MoveIt2 Servo* which utilizes PID controller and *ros2_control* for servoing, while a finite-state machine coordinates the handover sequence. A digital twin maintained within *MoveIt2* provides collision checking throughout execution.

1) *Real-Time Target Filtering and Prediction*: Object pose estimates are filtered before visual servoing to reject false detections and compensate for temporary tracking loss. First, a velocity gate removes outlier detections by evaluating the instantaneous target velocity,

$$\mathbf{v}_k = \frac{\mathbf{p}_k - \mathbf{p}_{k-1}}{\Delta t}. \quad (1)$$

Measurements producing $\|\mathbf{v}_k\| > 1.0$ m/s are rejected, while the previous valid pose is retained for up to five frames. Remaining poses are smoothed using a One-Euro filter [7] and extrapolated using a constant-velocity predictor,

$$\mathbf{p}_{\text{pred}} = \mathbf{p}_k + \mathbf{v}_{\text{est}}(\tau \Delta t), \quad (2)$$

where $\mathbf{v}_{\text{est}}$ is estimated over three frames and $\tau = 4$ is the prediction horizon.

2) *Adaptive Target Tracking*: During the approach phase, the robot continuously tracks the predicted object position. The desired tracking distance decreases as a function of target velocity,

$$d(t) = d(t - \delta t) - \left[\gamma_{\min} + (\gamma_{\max} - \gamma_{\min}) \frac{\|\mathbf{v}_{\text{est}}\|}{v_{\text{ref}}} \right] \delta t, \quad (3)$$

with $\gamma_{\min} = 0.05$ m/s, $\gamma_{\max} = 0.5$ m/s, and $v_{\text{ref}} = 0.5$ m/s. Commanded positions are generated through exponential interpolation,

$$\mathbf{p}_{\text{cmd},k} = \mathbf{p}_{\text{cmd},k-1} + \alpha(\mathbf{p}_{\text{pred},k} - \mathbf{p}_{\text{cmd},k-1}), \quad (4)$$

where α controls the tracking responsiveness.

3) *Waypoint Interpolation*: Motion between predefined robot poses is generated by independently interpolating Cartesian position and orientation:

$$\mathbf{x}(\beta) = \mathbf{x}_{\text{start}} + \beta(\mathbf{x}_{\text{goal}} - \mathbf{x}_{\text{start}}), \quad (5)$$

$$\mathbf{q}(\beta) = \text{SLERP}(\mathbf{q}_{\text{start}}, \mathbf{q}_{\text{goal}}, \beta). \quad (6)$$

The interpolation parameter $\beta = f(k/M)$ follows a configurable interpolation profile. Linear interpolation was used for transport motions.

4) *Kinematic Grasp Filtering*: Grasping initiates upon object velocity stabilization and target distance convergence. Before execution, candidate grasps are validated against the current robot configuration. Each grasp is first evaluated using inverse kinematics (IK), after which feasible solutions are assessed through the manipulator Jacobian,

$$J(q) = U\Sigma V^T, \quad (7)$$

where $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_6)$ contains the singular values. Candidates with $\sigma_{\min} < 0.1$ are discarded to avoid configurations near kinematic singularities. The remaining grasps are ranked according to the Jacobian condition number,

$$\kappa(J) = \frac{\sigma_{\max}}{\sigma_{\min}}, \quad (8)$$

favoring grasps with lower condition numbers and therefore greater Cartesian dexterity. Ties are resolved by selecting the candidate with the largest minimum singular value.

5) *Closed-Loop Tactile Grasping*: To grasp unknown objects safely, tactile measurements are filtered online and converted into binary contact events using a hysteresis threshold derived from an initial calibration phase:

$$P_{\text{high(low)},i} = \mu_i + k_{\text{high(low)}} \max(\sigma_i, \epsilon), \quad (9)$$

where $k_{\text{high}} = 16$, $k_{\text{low}} = 1$, and $\epsilon = 10^{-6}$. Position controller commands the gripper to close incrementally through precomputed joint positions until bilateral tactile contact is detected, after which a small object-dependent squeeze offset is applied to secure the grasp.

6) *Force-Torque Touchdown Detection*: During object placement, the wrist-mounted force-torque sensor detects contact with the table surface. A short calibration estimates the force bias immediately before descent. During placement, the filtered force magnitude

$$\|\mathbf{f}_{\text{net}}\| = \sqrt{(f_{k,x} - f_x)^2 + (f_{k,y} - f_y)^2 + (f_{k,z} - f_z)^2} \quad (10)$$

is monitored continuously. When $\|\mathbf{f}_{\text{net}}\|$ exceeds a predefined threshold, the downward motion is terminated and the gripper is released.

IV. BENCHMARKING AND EVALUATION

The qualification benchmark consisted of 288 known configurations evaluated under a repetition-permissive protocol with symmetric containers. The live finals comprised 24 previously unseen configurations with diverse object geometries, fillings, transparency, and deformability, and without trial repetitions.

TABLE I: RGMC 2026 results. Success rates are computed as successful trials over total trials, excluding hardware failures to ensure consistency with the finals evaluation protocol.

Phase	Success	Placement Err. (mm)	Score	Time (s)*	Rank
Qual.	75.9%	25.20	71.41	8.75	1st
Finals	75.0%	21.75	55.26	6.08	4th

*Measured from initial visual perception activation onward.

As summarized in Table I, the system achieved the highest qualification rank with a success rate of 75.9%. In the finals, the system maintained a 75.0% success rate despite increased environmental variability and a hardware failure the day prior that disabled one tactile sensor, reducing closed-loop grasp feedback. The small performance gap in this context suggests stable and consistent system behavior across both evaluation settings. Average handover execution time decreased from 8.75 s in qualification to 6.08 s in the finals, corresponding to a 30.5% reduction in interaction latency following post-qualification optimization and motion tuning.

V. CONCLUSION

This work presented a distributed robotic system for human-to-robot handover tasks using standard industrial hardware. The results demonstrate that safe, stable, and reactive performance can be achieved without specialized compliant grippers when sufficient sensing modalities are integrated with tightly coupled perception, object-agnostic grasp reasoning, and motion control. This is enabled through real-time feedback loops and compute-intensive processing that compensates for the lack of mechanical compliance.

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